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Palaeomagnetism of mid-Miocene leucitite volcanics in eastern Australia

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SUMMARY

The purpose of this study is to provide a new high-quality palaeomagnetic pole for mid-Miocene Australia using results from leucitite lavas in Eastern Australia, given their precise $^{40}Ar/^{39}Ar$ ages. We review published palaeomagnetic data for Australia during the mid-Miocene and examine palaeosecular variation during the 15–17 Ma interval. The average palaeomagnetic direction from leucitite lavas ($D = 190.2°$, $I = 57.1°$, $k = 33.9$, $a_{95} = 6.7°$) and the palaeomagnetic pole ($\lambda = -80.2°$, $\phi = 95.4°$, $K = 20.5$, $A_{95} = 8.6°$, $N = 15$, $Q = 5$) are statistically indistinguishable from early mid-Miocene poles from Nandewar Volcano and from the Port Campbell and Glenample Formations, demonstrating that older palaeomagnetic results can be robust despite outdated methodology (e.g. blanket demagnetization, few samples per site). We evaluate palaeosecular variation (PSV) during the mid-Miocene utilizing the combined data set ($N = 85/93$; cut-off), and calculate a mid-Miocene combined pole for Australia ($\lambda = -80.0°$, $\phi = 103.6°$, $K = 26.9$, $A_{95} = 3.0°$, $N = 85$, $Q = 6$). The pole yields a palaeolatitude estimate for the Cosgrove Hotspot of $\lambda_{15-17Ma} = 40.3°S \pm 3.0°$, consistent with the present-day inferred location of the Cosgrove Hotspot at 40.5°S. Scatter of virtual geomagnetic poles (VGPs) is estimated at $15.7°(^{17.4}_{14.1}$; 95 per cent limits), and is remarkably similar to the Model G and TK03 values from 0 to 5 Ma and 0 to 10 Ma lavas at comparable latitudes.

Key words: Palaeomagnetic secular variation; Palaeomagnetism; Australia; Hotspots.

1 INTRODUCTION

The process of reconstructing Earth palaeogeography for Jurassic and younger times relies strongly upon deciphering the record of marine seafloor magnetic anomalies, and upon palaeomagnetic poles for various continents to confirm such interpretations. Subduction of pre-Jurassic seafloor precludes the use of any marine magnetic anomaly records for older times, leaving palaeomagnetic poles as the sole basis for quantitative palaeogeographic reconstruction. Given a sequence of poles from a continental block through time, an apparent polar wander path (APWP) can be defined. From this path the continent's ancient latitude and orientation relative to the N–S oriented geomagnetic field lines can be determined (palaeolongitude cannot be directly constrained using palaeomagnetic methods). The shape of any given APWP is determined by the choice of palaeomagnetic poles used, and the method of interpolation used to fit data. Large time gaps between poles may obscure tectonic changes, and as such, a major goal of palaeomagnetism has been, and continues to be, the narrowing of these gaps with reliable poles.

It was to this end that, in the middle of last century, pioneering palaeomagnetists began studying the magnetism of Australian rocks (Green & Irving 1958; Irving & Green 1958; Irving *et al.* 1961). An early focus was on Cenozoic igneous rocks of Eastern Australia (Mumme 1963; Robertson 1966; Facer 1968; Wellman *et al.* 1969; McElhinny *et al.* 1974; Wellman 1975), and study of sediments (Bishop *et al.* 1982; Idnurm 1985) and laterites (Schmidt *et al.* 1976; Schmidt & Embleton 1976; Embleton 1981; Schmidt *et al.* 1982) followed. By the late 1980s a series of results had been published for the Australian Plate and although arguments persisted over details in the Cenozoic path of Australia (Idnurm 1990; Musgrave 1989, 1991), it seemed that a good representation of poles for Australia during the post-Cretaceous existed.

However, as the number of poles in global data sets grew, and as refined analytic techniques facilitated the reliable measurement of stable palaeomagnetic remanences from more diverse rock types and from deeper in geologic time, it became apparent that not all palaeomagnetic results were uniformly reliable. Remagnetization, incompletely removed overprints, uncorrected tectonic tilting, incorrect age assignments, and insufficient sampling affected some older, published palaeomagnetic data sets. It was no longer feasible to build plate models simply by indiscriminately compiling all the results for each plate, and selecting a method of fitting through the data. In response to this pressure, a schema of quality criteria (Q)

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for addressing the reliability of palaeomagnetic results was proposed (Van der Voo 1990). High-quality poles had well-constrained ages, documented magnetic mineralogy, detailed demagnetization and field tests to demonstrate primary magnetization.

The intended consequence of applying quality filter to palaeomagnetic data were to discharge any poles that may have suffered from aforementioned problems, leaving a fit only through demonstrably higher-quality data. While the use of more robust data are unquestionably desirable, quality filtering heralded a philosophical shift in the way palaeomagnetic data was appraised. The relative abundance of palaeomagnetic data and resultant increase in the scrutiny and filtering of this data set placed the onus on the modern researcher to demonstrate the *a priori* reliability of data. While it was stressed originally (and later reiterated) that the criteria schema proposed did not demonstrate that low Q poles were bad (Van der Voo 1990, 1993), the unintentional consequence remains that early studies, having been published before the many sources of palaeomagnetic complexity were widely recognized and before accepted solutions existed, are routinely disregarded.

The effects of this are evident in the pole choices made for Australia in modern plate reconstructions. There are 42 poles from the post-Cretaceous record of Australia listed in the recent compiled database (GPMDB v4.6b Pisarevsky 2011; McElhinny & Lock 1996), but the varied application of the Q-index has led to inconsistent and under-represented post-Cretaceous results for Australia in modern palaeomagnetic pole compilations (compare Besse & Courtillot 2002; Torsvik *et al.* 2008, 2012). The purpose of this study is to begin to address this shortcoming, and to compare modern results with earlier palaeomagnetic studies for this interval.

1.1 Geological background and sampling

Leucitite lavas in eastern Australia are exposed along a ∼630-km-long north–south linear chain from Byrock to Cosgrove (Birch 1976; Cundari 1973, 1989; Fig. 1). The rocks were generated by the Cosgrove Hotspot, that, along with the volcanos preserved in central Queensland form the longest continental hotspot chain in the world (∼2000 km; Davies *et al.* 2015; Sutherland 1981). Other hotspot tracks preserved on the Australian Plate from this time are the Eastern Australian hotspot track (located to the east of the Cosgrove track), as well as the Tasmantid and Lord Howe seamount chains.

Results of $^{40}Ar/^{39}Ar$ geochronology indicate that the Cosgrove hotspot was active from at least 34 Ma (Cape Hillsborough Beds) to 9 Ma (Cosgrove), with the linear arrangement of leucitite volcanic fields reflecting the path of the Australian Plate moving North over the Cosgrove hotspot during the 18–9 Ma interval (McQueen *et al.* 2007; Cohen *et al.* 2008, 2013). Most of the leucitite magmas were erupted between 15 and 17 Ma, contemporaneous with East Australian hotspot activity at Warrumbungle Volcano to the northeast, being ∼1.5 Ma younger than activity at Nandewar Volcano (northeast) and 2 Ma older than $^{40}Ar/^{39}Ar$ ages at Canobolas to the East (Cohen *et al.* 2008; Vasconcelos *et al.* 2008). Although leucitite flows younger than at the Cosgrove locality have not been found, a shield volcano, Macedon, lies to the southwest amongst magmas of the Newer Volcanic Province (NVP), and unlike the edge driven convection that produced NVP magmas, Macedon may be the result of the melting of Cosgrove Hotspot material (Davies & Rawlinson 2014). The present-day location of the Cosgrove Hotspot is inferred from finite frequency tomography to lie ∼300 km beneath northeastern Tasmania and the Bass Strait (Montelli *et al.* 2006; Whittaker *et al.* 2013).

The leucitite volcanics form low-lying (often <30 m thick), monogenetic flows presumed to have been erupted from vents or fissures during single events. The magmas erupted onto basement rocks: deformed Palaeozoic sediments and granitic rocks, and at some localities (e.g. Mt Bygalorie and Wilga Tank, El Capitan) upon Cenozoic sediments that unconformably overlie basement. The magmas are interpreted to have followed shallow depressions that have since eroded leaving the lava flows exposed as lobate hills and ridgelines.

The leucitite mineral assemblage in New South Wales (NSW) is characterized by iddingsitized olivine and less commonly by clinopyroxene phenocrysts set in a groundmass of clinopyroxene, leucite, titanomagnetite (and titanomaghemite), nepheline and mica. Sparse, microscopic pegmatoid veinlets have been commonly observed and are comprised of similar minerals with the exception of olivine, which is absent from these veins (Cundari 1973). All lavas along the Cosgrove track exhibit OIB-like trace element patterns (Davies *et al.* 2015). The leucitites are silica-undersaturated and, compared to their basaltic counterparts in Queensland, compositions have higher concentrations of Ti, Mg, K, P, Cr, Ni, Zn and trace elements, and lower Si, Al and Na concentrations. Isotopic data for the leucitites (Sr-Nd, Nd-Hf, Pb-Pb) indicate a metasomatized mantle source (EM-1/EM-2), with no evidence for crustal assimilation (Ewart 1982; Paul *et al.* 2005).

The total volume of leucitite lava is estimated at less than 3 km^3, a strikingly small magma volume when compared to the ∼10 000 km^3 magma volumes that comprise each central volcano of the northern Cosgrove Track (Wellman & McDougall 1974; Cundari 1989). Differences in composition and greater abundance of trace elements in leucitites compared to their basaltic counterparts in central Queensland have been attributed to greater depth of melting beneath the thicker lithosphere of central NSW compared to that of Queensland (Davies *et al.* 2015).

With the exception of oriented hand specimens collected at the Cosgrove locality (CG1), all samples were collected from outcrop with a gasoline powered, handheld Pomeroy drill. Multiple samples (from 8 to 12) were collected at 23 individual sites. Site location was recorded by handheld GPS, and sample orientation was taken by magnetic and solar compass, although prevalent overcast conditions during a large duration of fieldwork prevented solar measurements at all localities. Magnetic orientations were corrected using WMM2015-2015 (Chulliat *et al.* 2015). Specimens were prepared from samples by slicing cores into 2.1 cm lengths with a brass-blade rock saw, in preparation for demagnetization.

2 ROCK MAGNETIC ANALYSIS AND ANISOTROPY OF MAGNETIC SUSCEPTIBILITY (AMS)

2.1 IRM acquisition and curie balance

Small volumes of crushed rock were prepared from specimens at each site and loaded into a quartz glass holder enclosed by quartz wool for measurements on a Petersen Instruments Variable Field Translation Balance (VFTB). Isothermal remanent magnetization (IRM) was acquired at room temperature with increasing fields (to 500 mT) to identify the presence of magnetic phases of higher coercivity than titanomagnetite series minerals (>300 mT). Temperature-Magnetization heating and cooling curves were also acquired to further characterize carriers of magnetization in the

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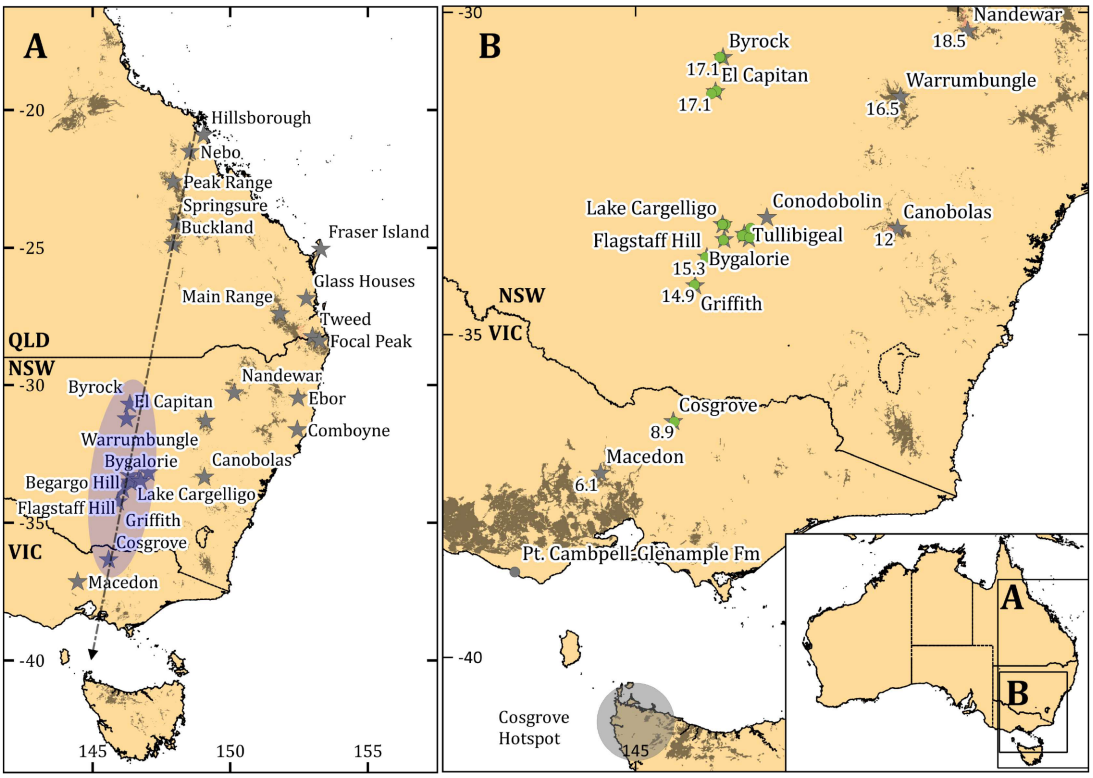


Figure 1. Map overview of (A) Central Volcanic provinces of Eastern Australia, with ellipse encompassing leucitite occurences. The dashed arrow represents the approximate path of the Cosgrove Hotspot. (B) Leucitite occurences with $^{40}Ar/^{39}Ar$ ages indicated (Ma) (Cohen *et al.* 2013, 2008; McQueen *et al.* 2007). Green dots indicate palaeoomagnetic sampling sites. Grey stars indicate hotspot derived volcanos. Grey circle indicates sampling locality for the Port Campbell-Glenample paleomagnetic study (Idnurm 1985). QLD: Queensland, NSW: New South Wales, VIC: Victoria.

leucitites (0–700 °C). Where such curves were irreversible, a second curve was acquired to confirm the nature of the newly formed magnetic phase (Fig. A1).

We found that samples show IRM saturation below 300 mT indicating little (if any) hematite present, although Curie balance curves for some sites were irreversible. Curie balance measurements were either reversible and indicative of titanomagnetite dominated magnetic assemblages, or irreversible, indicative of the transformation of maghemite to hematite upon heating (e.g. BK2-2, Fig. A1). The degree of maghemitization reflects post-eruptive alteration of the original igneous mineralogy, possibly by deuteric oxidation during cooling (Dunlop & Özdemir 2001). Regardless, the absence of hematite is evidenced by effective AF demagnetization and thermal demagnetization being complete by 600 °C.

2.2 Anisotropy of magnetic susceptibility

The purpose of measuring AMS ellipsoids was to examine the rock fabric and assess flow fabric preservation at each site, and to evaluate the possibility of tilting of the sampled flows. Bulk susceptibility (K_m) varies from 2.26×10^{-3} SI to 5.25×10^{-2} SI, with a mean of 1.38×10^{-2} SI across the entire data set, consistent with the volcanic nature of the rocks (Fig. A2). This range of values indicate that the measured fabrics are dominated by the ferromagnetic mineral assemblage. The degree of anisotropy of the magnetic susceptibility ellipsoid is low, as to be expected for undeformed volcanic rocks, with corrected anisotropy factor (P_J) values across the entire data sett ranging from 1.003 to 1.047, with a mean of 1.017. This low degree of anisotropy is reflected within each site as well (Fig. A2). The shape of the AMS ellipsoids are variable between specimens and range from very prolate ($T = –0.891$; K_1–K_3) to very oblate ($T = 0.967$; K_1–K_2), and while sites BH1 and CG1 exhibit a dominant oblate fabric, there is considerable variation in shape parameter within sites (Fig. A2b). No systematic shape fabric is evident within the leucitites as a whole (mean $T = 0.118$, $\sigma = 0.403$) (Fig. A2).

At the site level, principal axes directions of the AMS ellipsoid often cluster (Fig. A2c), with steeply inclined (>50°) K_3 axes (with the exception of TB3 and EC3) and shallow inclination K_1 and K_2 axes (in geographic coordinates) (Fig. A2). K_1 and K_2 axes cluster as elongated ellipses along a shallowly inclined girdle. This pattern however, does not hold for all analyses from a given site, or for that matter, for all sites. Examples of inverse AMS fabrics are present (e.g. site EC3 has a steep K_2 axis and shallow K_1 and K_3 axis) and within sites individual specimens occasionally show inverse fabrics (K_1-K_2-K_3 substitution), a phenomenon reported from studies of other undeformed mafic rocks where single domain magnetite contributions dominate the AMS signal (Tarling & Hrouda 1993). The AMS signal could also have been subject to erasure and replacement if the flow settled before solidifying (e.g. convection and/or gravity settling). Complication of any primary fabric by secondary fabrics produced during maghemitization (which would be controlled by microfracture orientation) and erasure by cooling stresses (local residual stresses present due to temperature gradients during cooling) is also possible (Rochette *et al.* 1999). Furthermore, it is reasonable to presume that some measurements could be a result of a mixture of multiple and single domain magnetite grains in relative quantities such that intermediate AMS fabrics result.

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Table 1. Site mean Fisher's statistics for eastern Australian leucitite lavas. Dec: Declination, Inc: Inclination, k: precision parameter (directions), a_{95}: 95 per cent confidence angle (directions), n: number of samples, A_{95}: 95 per cent confidence angle (VGPs). ∗Site excluded from pole calculation (see text).

Sample	Dec (°)	Inc (°)	k	a_{95} (°)	n	Site Latitude (°N)	Site Longitude (°E)	VGP Latitude (°N)	VGP Longitude (°E)	A_{95} (°)	Polarity Chron
CG1	170.2	47.2	419.22	2.99	6	–36.351	145.599	–78.5	276.8	3.12	C4.r.2r/C4Ar.1r
GQ1	–	–	–	–	-	-	-	-	-	-	-
GQ2	–	–	-	-	-	-	-	-	-	-	-
BH1	–	–	-	-	-	-	-	-	-	-	-
BH2∗	165.1	69.9	9.44	21.76	6.5	-33.527	146.361	-67.2	169.5	34.65	C5Br/C5Bn.1r
FH1	168.1	61.6	83.28	6.73	6	-33.793	146.087	-77.1	188.7	9.13	C5Br
LC1	180.3	54.6	29.85	9.01	9	-33.285	146.388	-88.1	138.8	10.69	C5Br/C5Bn.1r
LC2	193.4	54.5	32.01	11.38	6.5	-33.285	146.348	-78.8	69.0	13.47	C5Br/C5Bn.1r
LC3	189.1	45.4	32.77	11.88	6	-33.282	146.344	-79.9	19.6	12.03	C5Br/C5Bn.1r
LC4	164.2	56	62.64	8.7	5	-33.282	146.344	-76.7	217.7	10.59	C5Br/C5Bn.1r
LC5	187.2	51.9	62.24	7.37	7.5	-33.286	146.345	-83.9	51.1	8.32	C5Br/C5Bn.1r
TB1	169	69.5	66.38	7.84	6.5	-33.444	146.673	-68.7	165.0	12.40	C5Br/C5Bn.1r
TB2	173.8	69.4	46.89	9.88	6	-33.463	146.654	-69.9	157.5	15.59	C5Br/C5Bn.1r
TB3	197.5	66.5	132.11	5.85	6	-33.46	146.653	-69.7	111.9	8.73	C5Br/C5Bn.1r
TB4	174.1	51.5	37.56	10.48	6.5	-33.493	146.783	-84.9	250.3	11.75	C5Br/C5Bn.1r
TB5	201.5	36.4	76.12	7.73	6	-33.486	146.768	-66.8	27.4	6.88	C5Br/C5Bn.1r
TB6	222.9	48.9	45.36	9.52	6.5	-33.345	146.785	-53.5	63.2	10.21	C5Br/C5Bn.1r
EC1	-	-	-	-	-	-	-	-	-	-	-
EC2	1.3	-49.4	41.03	8.81	7	-31.217	146.251	88.5	195.9	9.53	C5En/D5Dr.1r
EC3∗	209.9	26.6	2.75	44.15	6.5	-31.259	146.176	-57.7	30.7	35.26	-
BK1	221.3	64.5	128.95	6.3	5.5	-30.706	146.327	-54.6	94.4	9.04	C5Cr
BK2	-	-	-	-	-	-	-	-	-	-	-
BK3	216.3	55.4	30.17	12.39	6	-30.693	146.307	-59.4	76.0	14.92	C5Cr

3 PALAEOMAGNETIC ANALYSIS

3.1 Demagnetization

Demagnetization and rock magnetic measurements were carried out at the Curtin University Palaeomagnetic Laboratory. Magnetic remanence was measured using an Agico JR6 Spinner Magnetometer in a field free room made of mu-metal. Progressive AF and Thermal demagnetization (at least 17 steps per specimen) was employed using a tumbling AF demagnetizer (maximum field 100mT) and an ASC magnetically shielded furnace (maximum temperature 700 °C). Component magnetizations were determined by principal component analysis using at least four demagnetization steps with maximum angular deviation (MAD) $<10°$ (Zijderveld 1967; Kirschvink 1980) and in cases where simultaneous removal of magnetic components occurred, by the arc method ($n \geq 4$, MAD $<15°$; Zijderveld 1967; Kirschvink 1980; McFadden & McElhinny 1988) using PalaeoMag software Jones (2002). Site mean directions were determined by Fisher statistics Fisher (1953), assigning a 0.5 weighting to directions fit by the arc method. Virtual Geomagnetic Poles (VGPs) were calculated using site mean directions, and paleomagnetic poles were calculated as Fisher means of VGPs calculated from individual sites. Further tests and calculations were facilitated by the use and expansion of select tools in the pmagpy catalogue Tauxe *et al.* (2016). Site mean results are given in Table 1 and all fits are described in Supplementary Information.

3.2 NRM and magnetic components

Natural remanent magnetizations (NRMs) showed no preferred directional clustering, and intensities are skewed from a median intensity of 1.97 A m^{-1} towards higher intensities (interquartile range: 0.71–8.32 A m^{-1}). This median is slightly higher than expected intensity for basalts (∼1 A m^{-1}) and high intensities are probably the result of lightning IRM overprints as discussed in detail below.

Most samples from sites exhibited stable demagnetization (21/23 sites), the exception being Griffith Quarry sites GQ1 and GQ2. Example demagnetizations are shown in Fig. 2. Directions were consistent and well-clustered ($k > 30$) for 16 of the 23 sites sampled (Table 1, Fig. 3). Progressive demagnetization revealed one to three components of magnetization; a low coercivity component (s1) unblocked at low temperature <200 °C or with peak fields of <20 mT, revealing either a high coercivity characteristic component of magnetization (c1), or an intermediate coercivity overprint along with a characteristic component (s2, c1), or simultaneously demagnetizing two high coercivity components, necessitating the use of demagnetization arcs (c1a). A small number of samples showed only a single component of magnetization (either as the result of sampling quarry faces, or remagnetization by lightning). Comparison of AF and thermal demagnetization showed no obvious difference in resolving individual palaeomagnetic components, and with AF demagnetization proving effective this method was chosen as treatment for the majority of samples.

As the lava flows commonly occur as slightly elevated topography relative to the surrounding flat plains we suspected both the overprint (s1, s2) components and simultaneous demagnetization of higher coercivity components (c1a) to be the result of lightning IRM overprinting a primary TRM. The direction of lightning IRM should be random (Watson 1956), so we used a randomness test to check for lightning-induced directions. In contrast, characteristic remanences that have a primary origin are expected to have non-random, well-clustered directions. Overprint component s1 is random for the majority of sites (16/21: 95 per cent confidence, 19/21: 99 per cent confidence). Secondary components within sites BH1, BH2, EC1, EC2 and LC5 are not random at the 95 per cent

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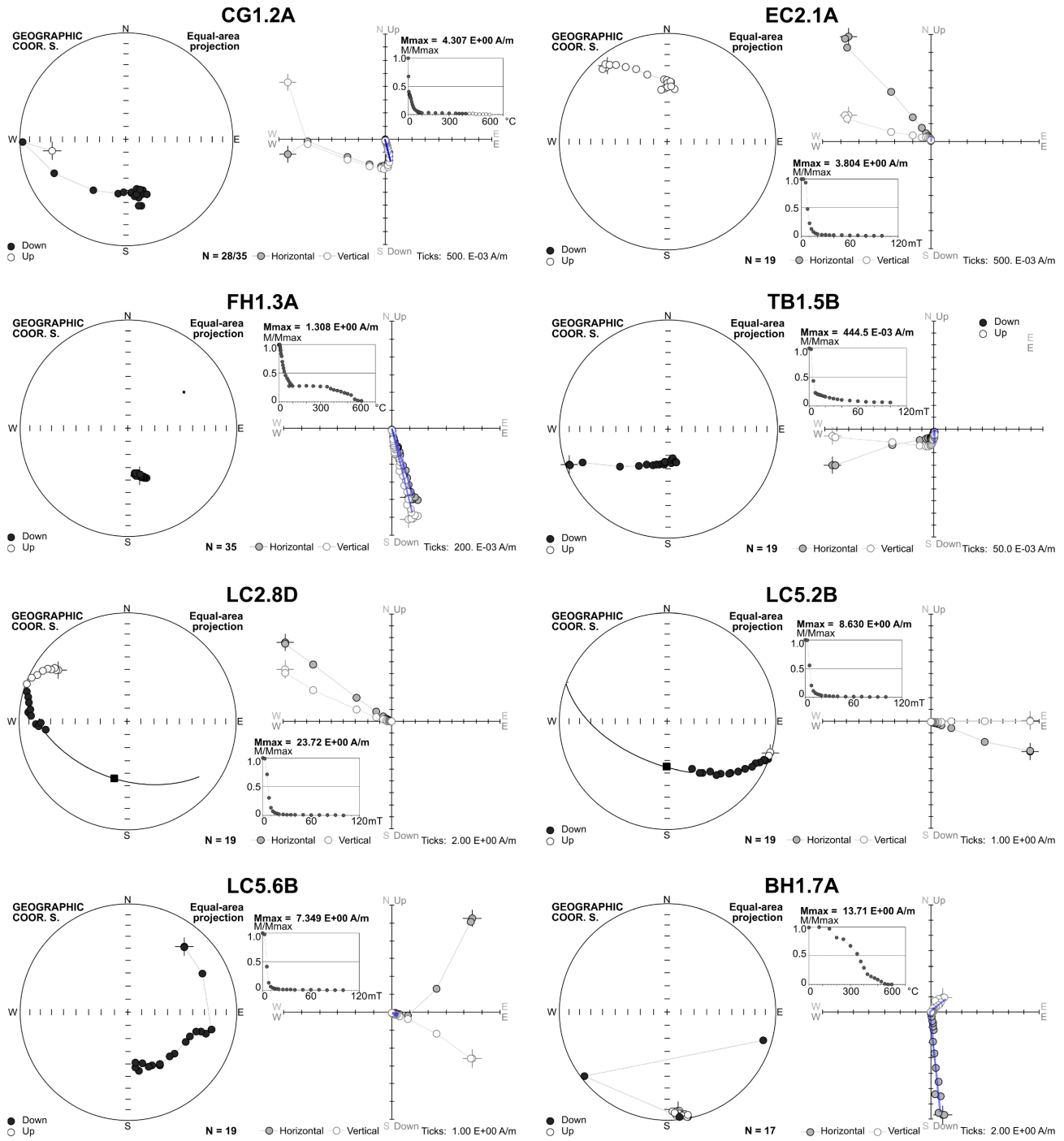


Figure 2. Orthogonal vector diagrams of typical demagnetization behavior (blue lines on orthographic plots show linear fits to characteristic magnetization). LC2.8D and LC5.2B show arc fits (black line: c1a) with squares indicating best fit along arc. BH1.7A depicts a sample inferred to be lightning remagnetized (c1L). *N* indicates number of stable demagnetization steps (shown).

confidence level, but only within sites EC1 and LC5 are they not random at 99 per cent confidence. Even so, s1 directions at these sites are incredibly scattered with precision parameters (k) of 2.78 (EC1) and 1.92 (LC5). Also, it suggests that differential overprinting by lightning is likely responsible for cases of higher coercivity overprints (s2) and the simultaneous removal of higher coercivity components in some samples (c1a).

Conversely, characteristic components (c1 and c1a) are grouped (not random: Watson 1956) for the majority of sites (18/21 sites: 95 per cent confidence, 16/21: 99 per cent confidence), these directions are interpreted as primary, reflecting the acquisition of magnetization upon cooling (i.e. magnetization). The only sites that reject the null hypothesis at 95 per cent confidence and not at 99 per cent are EC3 ($k = 2.75$) and BH2 ($k = 9.44$), and both have low precision. While we are not confident in assigning a polarity to the scatter of directions at EC3, BH2 is interpreted to be of reversed polarity although the large, within-site scatter precludes the use of this site in calculating a paleomagnetic pole. Sites that failed to

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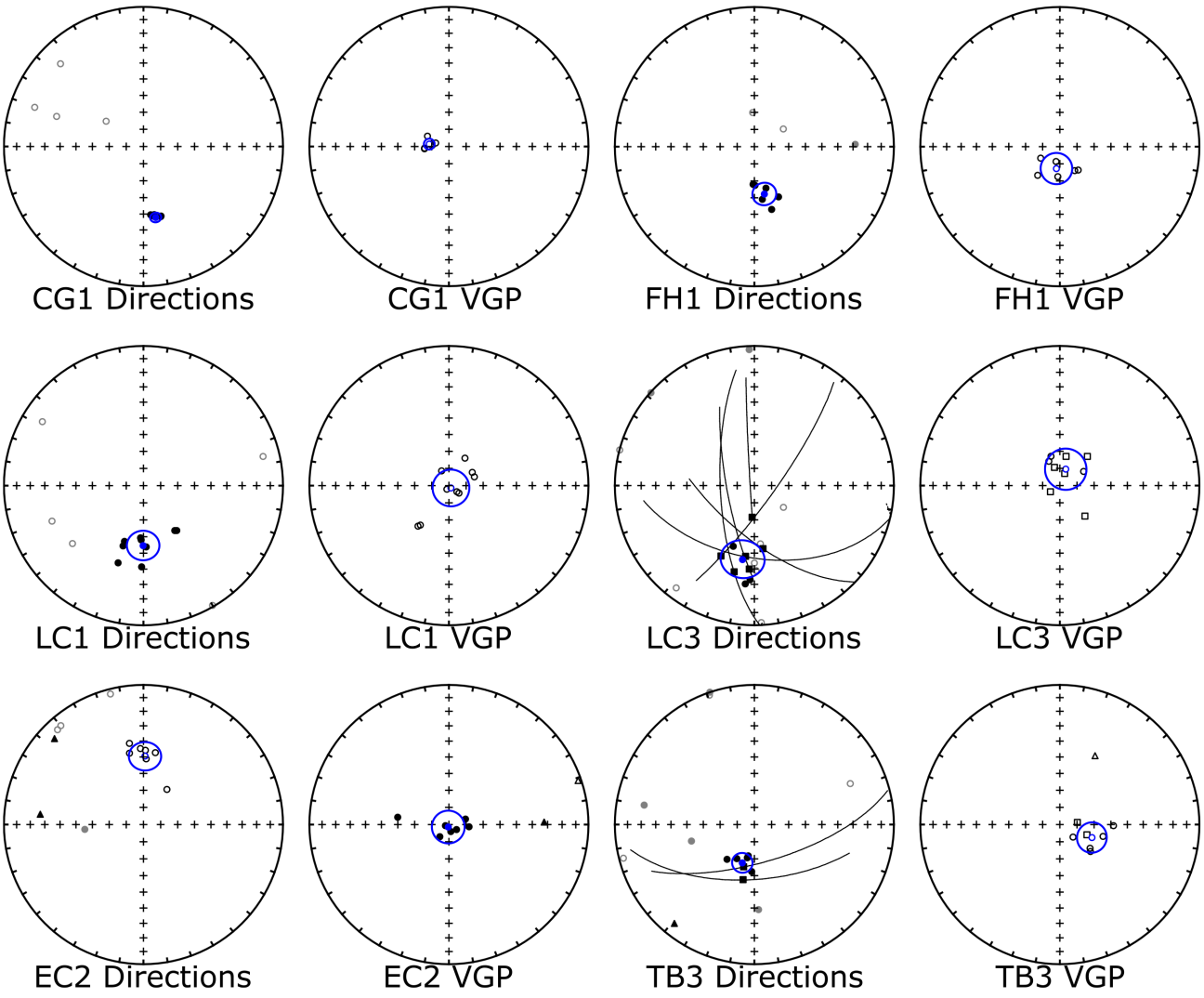


Figure 3. Representative equal area stereonets of directional components and VGPs by site. Filled Symbols indicate positive (downward) inclination directions and North Hemisphere VGP latitudes. Open symbols indicate negative (upward) inclination directions and South Hemisphere VGP latitudes. Grey symbols indicate overprint directions (s1, s2), Black circles/squares indicate directions inferred to be primary determined by line/arc fitting respectively (c1/c1a). Arc segments for such fits are dotted in upper hemisphere and solid in lower hemisphere. Black triangles indicate directions inferred to reflect lightning remagnetization (c1L). Fisher means for both directions and their VGPs (with a_{95} and A_{95}, respectively) are plotted in blue. Remaining sites illustrated in Figs A3–A4.

reject being drawn from a random distribution (at 99 per cent confidence) are considered to be dominated by lightning IRM (fits from samples suspected of lightning remagneization are given the label c1L). These samples were excluded from further consideration and site mean calculations. Some samples are completely remagnetized by lightning, even in sites where other samples preserve a primary magnetization (9/131 samples). These lightning remagnetized samples displayed a single component demagnetization to the origin not consistent with the direction of other samples at the site, and often had significantly higher NRM intensities than other samples from the same site.

3.3 Polarity, field tests and sampling palaeosecular variation

All of the sites with interpreted primary magnetization ($N = 16$) are of reversed polarity, with the exception of EC2, which is of normal polarity. Combination of $^{40}Ar/^{39}Ar$ ages (with 2σ errors) indicate that polarity determinations are compatible with the magnetic polarity time scale (Ogg 2012; Fig. 4). El Capitan site EC2 erupted either in chron C5En or D5Dr.1r, sites at Byrock (BK) coincide with chron C5Cr, the flow at Flagstaff Hill (FH1) erupted during reversed chron C5Br, and the remainder also fall within chron C5Br or C5Bn.1r. That many measured VGPs could come from within chron C5Br is unsurprising because its ∼0.8 Ma duration is relatively long compared with the mid-Miocene average of ∼0.2 Ma (Ogg 2012). The age of lava at Cosgrove (CG) falls within a short-normal polarity chron (C4An) although 2σ upper and lower error limits coincide, respectively, with upper and lower reversal boundaries of chron C4An. The reverse polarity recorded here could reflect either one of the adjacent reversed chrons (C4r.2r, or C4Ar.1r).

Characteristic magnetization at all sites are well determined, with $k > 30$ and $a_{95} < 15^\circ$. Notably, a small outcrop of baked sediment at the base of a lava flow was sampled (LC4), carrying a reversed magnetization ($D = 164.2^\circ$, $I = 56^\circ$, $k = 62.64$, $a_{95} = 8.7^\circ$, $N = 5$) with similar direction to the overlying lava, but the absence of any

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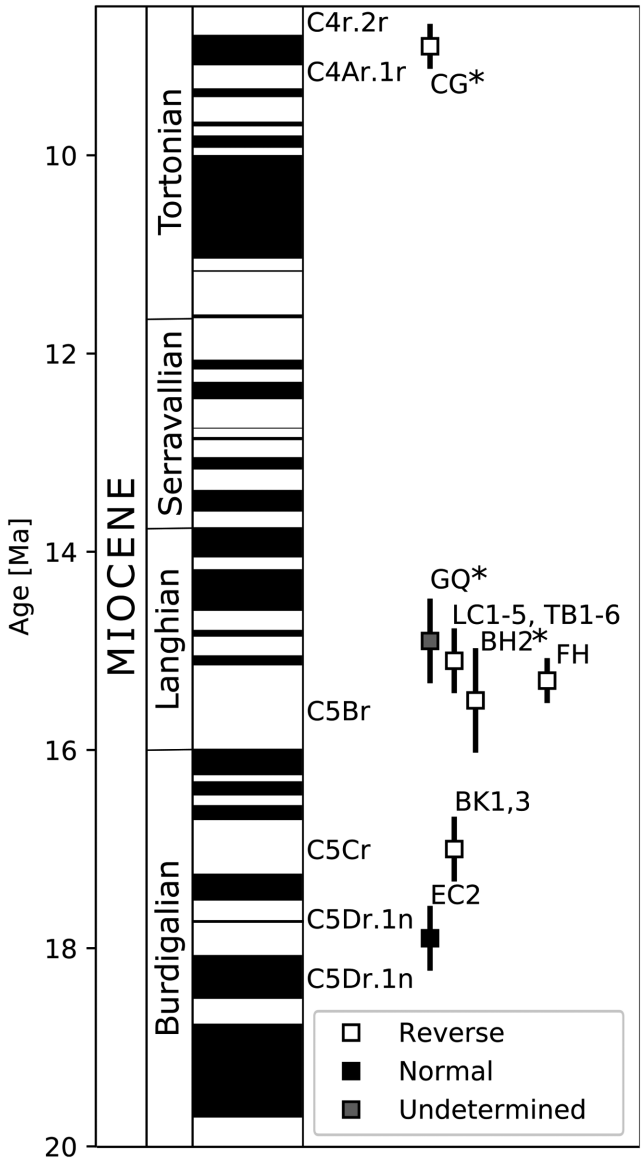


Figure 4. Polarity comparison of leucitite lava flows and the Geomagnetic Polarity Time Scale Ogg (2012). Possible polarity chrons for lavas are indicated. Sites marked with an asterisk were not included in pole calculation (see text). CG: Cosgrove, GQ: Griffith Quarry, FH: Flagstaff Hill, BH: Begargo Hill, LC: Lake Cargelligo, TB: Tullibigeal, EC: El Capitan, BK: Byrock.

surrounding country rock outcrop prevented a complete contact test. In most locations the basement upon which these rocks were erupted is not exposed, preventing extensive use of the contact test. Despite this, McQueen *et al.* (2007) reported dual-polarity remanence from a paleosol section beneath a reversely magnetized leucitite lava at Wilga Tank (El Capitan region) with VGPs that are distinct from the pole calculated for leucitite lavas in this study by both Watson's V test (V: 8.0, V_{crit} = 7.0) (Tauxe *et al.* 1991; Watson 1983) and McFadden & McElhinny's (1990) common mean test (Angle = 10.7°, Critical angle = 10.1°). The paleomagnetic direction for the paleosol is steeper, and therefore older, since Australia has been moving northward through the Cenozoic, suggesting preservation of magnetizations older than leucitite volcanism in surrounding country rock. Their result suggests that, at least in the El Capitan region, regional remagnetisation has not taken place. While there is no field test supporting a primary magnetization for the lavas in our study (positive contact test), magnetization directions are compatible with other mid-Miocene poles from Australia during this time (Glenample and Port Campbell Formations, Hawkesbury Sandstone, Nandewar Volcano) and there is no evidence of regional remagnetization of rocks through the region since the eruption of leucitites.

4 MID-MIOCENE PALAEOMAGNETIC POLE FOR EASTERN AUSTRALIAN

4.1 palaeomagnetic pole from leucitite lavas

Fisher means of characteristic magnetizations for well defined sites (Table 1) were used to calculate individual VGPs from which a palaeomagnetic pole for eastern Australian leucitites was determined. We exclude the Cosgrove locality's VGP (CG) from the pole calculation because of its much younger age (8.9 ± 0.2 Ma; Cohen *et al.* 2008) compared with the age from the remainder of sites (15–17 Ma). Including the Cosgrove site result would change the pole location by less than 2° in spite of the 6–8 Ma age difference from the other, older sites. The average palaeomagnetic direction for all sites (excluding Cosgrove) is D = 190.2°, I = 57.1°, (k = 33.9, a_{95} = 6.7°, N = 15). The Fisher mean of VGPs indicates a pole position for Australia of λ = −80.2°, ϕ = 95.4° (K = 20.5, S = 18.1°, A_{95} = 8.6°, N = 15) (Figs 5a and b). We assign a magnetization age for this pole of 15–17 Ma on the basis of published $^{40}Ar/^{39}Ar$ ages. We assign the pole a Q score of 5 (or 6 if the reversals in palaeosol beneath the Wilga Tank Leucitite McQueen *et al.* (2007) are considered as a positive contact test) to this pole. The lack of normal polarity flows (only one normal polarity flow sampled) prevent a meaningful reversal test (category 6) and poorly exposed basal contacts prevented *sensu stricto* field tests for baked contacts (Q-category 4).

The angular dispersion of VGPs of 18.1° ($^{14.2}_{24.2}$ 95 per cent confidence bounds) is typical for intermediate latitude sites indicating that palaeosecular variation (PSV) has been sampled, and that the geomagnetic field has been adequately represented. This conclusion is strengthened by a continuum of $^{40}Ar/^{39}Ar$ ages, and field evidence indicating individual eruptions from solitary vents. Also, as discussed in the following section, VGPs from this pole are statistically indistinguishable with other Australian poles of this age, indicating the robustness and consistency of mid-Miocene results from Australia.

4.2 Pole comparison and combined mid-Miocene pole for Australia

We conducted common mean tests using Watson's V statistic and McFadden & McElhinny's (1990) test to determine if VGP distributions from published Australian poles (Table 2) could have been drawn from the same underlying distribution as results from leucitite lavas (and each other) (Watson 1983; McFadden & McElhinny 1990; Tauxe *et al.* 1991). We found that VGP distributions for Nandewar Volcano (Wellman *et al.* 1969), Glenample and Port Campbell Formations (Idnurm 1985) and leucitite lavas are statistically indistinguishable. As discussed by Musgrave (1989), the original filtering method used to identify transitional VGPs at Nandewar Volcano is biased because transitional sites are not randomly distributed, so we test using the more robust unfiltered data set and later apply a variable VGP cutoff to robustly estimate VGP

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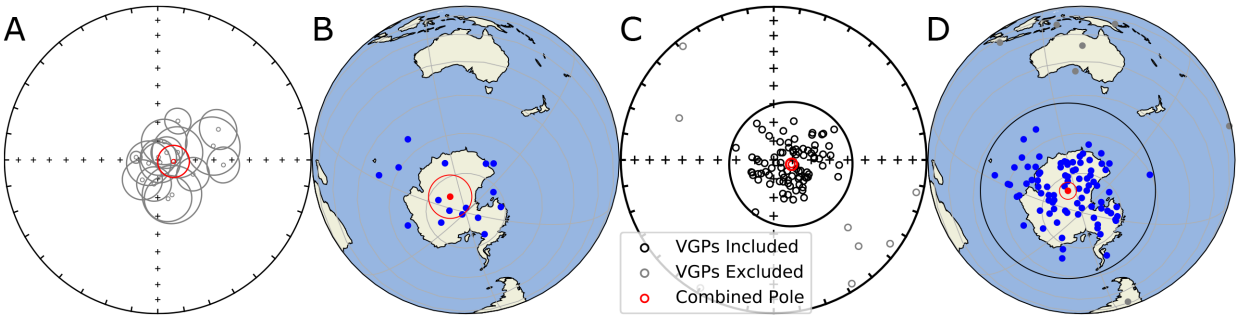


Figure 5. [A] Equal area stereonet of site mean VGPs (grey) and palaeomagnetic pole (red) for leucitite lavas, [B] Orthographic projection of leucitite VGPs (blue) and mean (red), [C] VGP distribution for combined mid-Miocene pole, with black circle indicating limits of Vandamme (1994) cut-off [D] Orthographic projection of combined VGP data (included: blue, excluded: grey), mean (red) and Vandamme (1994) cut-off (black).

Table 2. Mid-Miocene palaeomagnetic poles (North) for Australia (λ = Pole Latitude, ϕ = Pole Longitude), with associated precision parameter (K) and confidence angle (A_{95}) after variable VGP cut-off has been applied Vandamme (1994). N indicates number of sites included in pole, while Vandamme column indicates number of sites identified as transitional by filter. Q after Van der Voo (1990). Age determinations by a: $^{40}Ar/^{39}Ar$ ages, b: biostratigraphic ages (foraminifera), c: K-Ar ages, d: magnetization age presumed from magnetic direction. i: included in mid-Miocene combined pole.

Name	λ (°N)	ϕ (°E)	A_{95} (°)	K	N	Age (Ma)	Q	Reference	Vandamme
Leucitite Pole (this study)	80.2	275.4	8.6	20.5	15	15–17[a]	5	this study	0 [i]
mid-Miocene Combined Pole (this study)	80.0	283.6	3.0	26.9	85	~12–17	6	this study	
Glenample Fm./Port Campbell Lst.	77.2	303.5	3.9	40.7	30	~12[b]	6	Idnurm (1985)	0 [i]
Nandewar Volcano	79.7	257.5	4.5	26.5	40	~17 [c]	5	Wellman *et al.* (1969)	-8 [i]
Hematized Hawkesbury Sandstone	82.9	294.4	2.5	281.3	13	~15	3 [d]	Bishop *et al.* (1982)	0
Wilga Tank palaeosol	78.5	323.2	3.2	69.5	30	>17	4 [d]	McQueen *et al.* (2007)	-5
Warrumbungle Volcano	78.7	320.9	17.2	8	11	~14–16 [c]	4	McElhinny *et al.* (1974)	-1

dispersion (Wellman *et al.* 1969; Vandamme 1994). VGPs from Warrumbungle Volcano (McElhinny *et al.* 1974) are highly scattered [$K = 8$ after Vandamme (1994) filtering removing one site] so while common mean tests indicate each VGP sample could being drawn from the same VGP population, the McFadden & McElhinny (1990) classification is indeterminate. It is not clear why such high scatter is present at Warrumbungle Volcano, but it could be due to incompletely removed overprints, or a preponderance of transitional VGPs combined with limited sampling ($N = 12$ sites). Given this ambiguity we do not include VGPs from Warrumbungle Volcano in the combined pole and do not consider it in further analysis.

The pole from the Lapstone Monocline in the Hawkesbury Sandstone (Bishop *et al.* 1982) is not included in combined analysis because the magnetization is secondary, and the age of its magnetization is presumed from the pole position along Australia's APWP. Although hematite magnetization in these beds may have been acquired contemporaneously with leucitite and Nandewar Volcano magmatism the dependence of its age upon its palaeomagnetic direction would invoke circular reasoning. We also disqualify the Wilga Tankpalaeoosol because its age is presumed from its position along Australia's APWP, and as discussed earlier, the results are distinct from leucitite VGPs and possibly older (McQueen *et al.* 2007).

Given the consistency of both age and VGPs from leucitite lavas, Nandewar Volcano, and Glenample and Port Campbell Formations, it is acceptable to combine them, thereby improving the precision of the palaeomagnetic pole for Australia and facilitating a meaningful study of mid-Miocene PSV. Site mean VGPs were obtained from the aforementioned poles, and we applied a variable cutoff (Vandamme 1994) to deconvolve transitional directions from those reflecting secular variation of a stable geomagnetic field (Fig. 5c).

Only eight sites were recognized as preserving transitional field directions (all from Nandewar Volcano). The combined pole position is $\lambda = -80.0°$, $\phi = 103.6°$, with $K = 26.9$, $S = 15.7°$, $A_{95} = 3.0°$, $N = 85$ (Figs 5c and d). Including transitional directions does not significantly alter the pole position ($\Delta 1.9°$). The combined pole has a Q-score of 6, because although validated by step-wise demagnetization and principal component analysis from both leucitite lavas and Port Campbell and Glenample Formations, Nandewar Volcano results were blanket demagnetized (violating Q-category 3 in a strict sense). In spite of this, the combined pole both contains reversals, and passes a reversals test (C-class; 5° between means, 10.1° critical angle), a test that was not possible on leucitite lavas alone (McFadden & McElhinny 1990). A quantile–quantile test of the VGP data is consistent with a Fisherian distribution, VGP longitudes are uniformly distributed and VGP latitudes are exponentially distributed ($M_e = 0.818$, $M_u = 0.968$ (critical values = 1.094, 1.207).

5 MID-MIOCENE PALAEOSECULAR VARIATION

Our new integration of the mid-Miocene palaeomagnetic results from Australia allows us to examine the reconstructed field and characterize palaeosecular variation at this time. Fig. 6 shows the angular dispersion as a function of latitude (with 95 per cent confidence bounds) for individual studies, as well as for the combined pole result. The Vandamme (1994) variable cut-off was applied to site mean VGP populations for each pole, with the only exclusions being for the eight transitional results of Nandewar Volcano previously described, and for five results from the Wilga Tank Palaeosol. The sedimentary poles of the Glenample and Port Campbell Formation (Idnurm, 1985), Wilga Tank paleosol (McQueen *et al.* 2007) and Hematized Hawkesbury Sandstone (Bishop *et al.* 1982) all show

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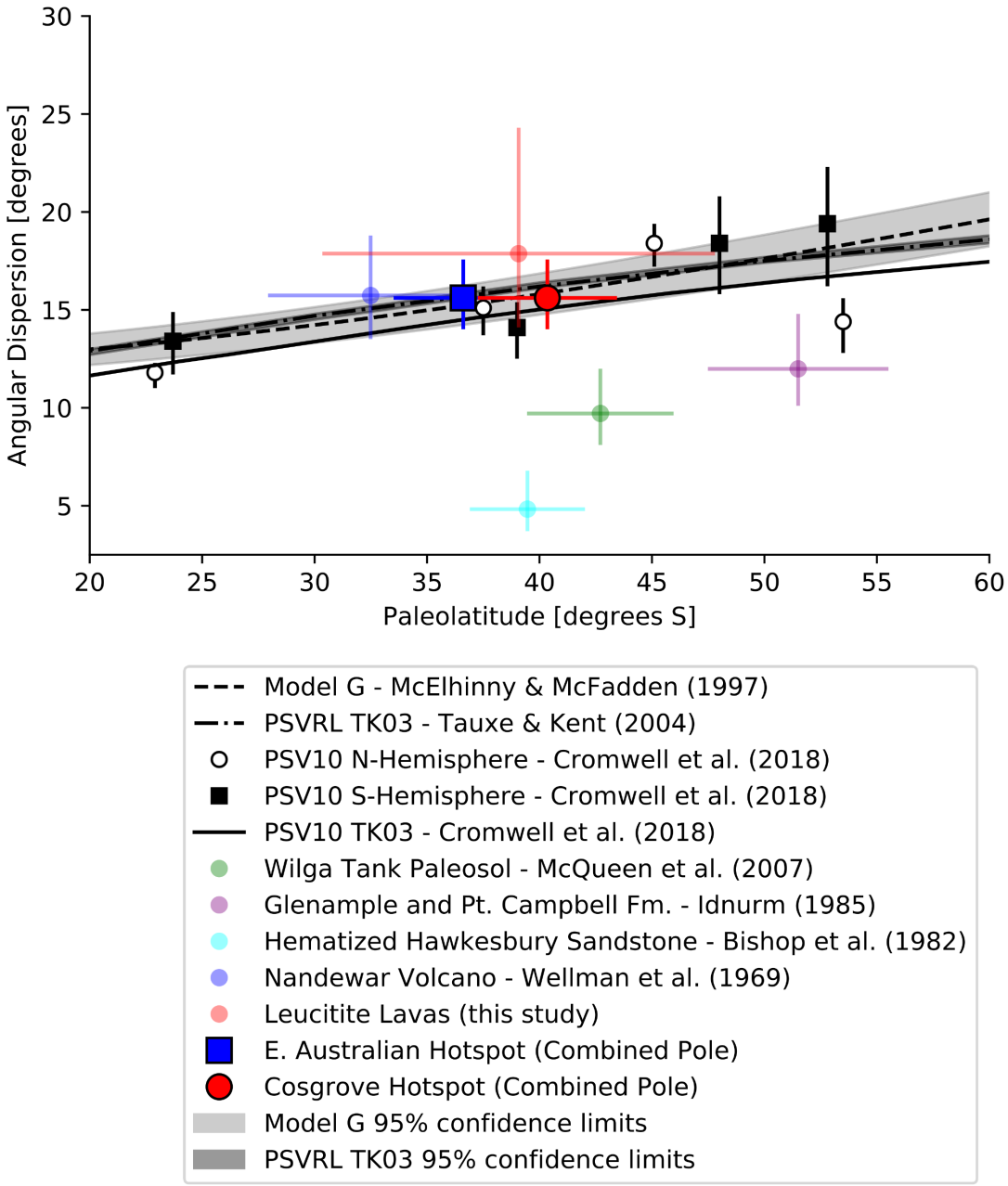


Figure 6. Angular dispersion with palaeoolatitude for mid-Miocene palaeomagnetic results from Australia compared with field models derived from databases of recent lavas (0–5 Ma lavas of the PSVRL data base: Model G, and TK03 McElhinny & McFadden (1997); Tauxe & Kent (2004), and 0–10 Ma lavas from the PSV10$_{vcut}$ database: TK03 Cromwell *et al.* (2018). White circles and black squares illustrate data from the PSV10$_{vcut}$ data base from the Northern and Southern Hemispheres, respectively.

lower than expected scatter, which is probably a result of acquisition of magnetization over longer intervals within samples themselves, so rather than recording an instantaneous measurement of the geocentric axial dipole (as is the case for TRM from volcanic results), some averaging of palaeosecular variation has already occurred within sites. For the Hematized Hawkesbury Sandstone, the effect is particularly pronounced, and the effect of PSV averaging within samples was discussed in detail by the original authors. The angular dispersion is slightly higher than expected for the leucitite lavas and marginally so for Nandewar Volcano (Wellman *et al.* 1969), but confidence limits overlap with model G and TK03 predicted values in both cases (McElhinny & McFadden 1997; Tauxe & Kent 2004; Cromwell *et al.* 2018). The robustness of the combined pole result ($N = 85$) is evident here, and yields a VGP scatter of 15.7°, ($^{17.4}_{14.1}$ 95 per cent confidence limits), that is remarkably consistent with expected angular dispersion for middle latitudes based upon both 0–5 Ma and 0–10 Ma lavas (Model G and TK03 fits for both PSVRL and PSV10$_{vcut}$ databases) (McElhinny & McFadden 1997; Tauxe & Kent 2004; Cromwell *et al.* 2018). Confidence intervals for the combined poles also overlap with limits of PSV10$_{vcut}$ data between 35° and 45° latitude for both hemispheres (Cromwell *et al.* 2018). On the basis of this consistency we conclude that geomagnetic field scatter from mid-Miocene Australian results was similar to that during the 0–10 Ma interval.

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6 CONCLUSIONS

We have presented new palaeomagnetic data for leucitite lavas in eastern Australia, and compared results with previous paleomagnetic data from Australia during the mid-Miocene. Good agreement between the results obtained in our study with modern techniques, and those of previous researchers, indicates that older paleomagnetic results can still be reliable therefore should not be disregarded simply on the basis of their antiquity alone. For the mid-Miocene we have presented a case for those Australian data that should be included in Cenozoic plate reconstructions. We have combined mid-Miocene VGP data where appropriate to provide a robust and precise estimate of the pole position of Australia during mid-Miocene times. Lastly, we show that paleosecular variation from Australian mid-Miocene rocks are similar to expected values from field models based on recent paleomagnetic data (0–5 Ma and 0–10 Ma data sets).

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SUPPORTING INFORMATION

Supplementary data are available at *GJI* online.

Figure S1. Representative IRM acquisition curves and Curie balance measurements for leucitites. Heating curve is represented by the solid line, and cooling by the dashed line. For sample BK2-2 the black and grey curie curves represent first and second heating, respectively. M: Magnetization, normalized to one (IRM) or initial value (curie balance).

Figure S2. Site Mean averages for multiple samples from individual sites (2σ errors). Corrected degree of anisotropy (P_J), Shape Parameter (T), and equal area projections of site mean AMS Fabric principal axes, indicating a steeply inclined minimum axis (K_3) and shallow girdle of K_1–K_2 axes.

Figure S3. Equal area stereonets of directional components and VGPs by site. See Fig. A4 for symbology.

Figure S4. Equal area stereonets of directional components and VGPs by site. Filled Symbols indicate positive (downward) inclination directions and North Hemisphere VGP latitudes. Open symbols indicate negative (upward) inclination directions and South Hemisphere VGP latitudes. Grey symbols indicate overprint directions (s1, s2), Black circles/squares indicate directions inferred to be primary determined by line/arc fitting, respectively (c1/c1a). Arc segments for such fits are dotted in upper hemisphere and solid in lower hemisphere. Black triangles indicate directions inferred to reflect lightning remagnetization (c1L). Fisher means for both directions and their VGPs (with a_{95} and A_{95}, respectively) are plotted in blue.

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